Question #02:

Data:

$C_1 = 10 \mu F = 10 \times 10^{-6} F$
$C_2 = 50 \mu F = 50 \times 10^{-6} F$
$C_3 = 25 \mu F = 25 \times 10^{-6} F$
$V = 250 V$

i) $q_1 = ?$, $q_2 = ?$, $q_3 = ?$

ii) $C_{Eq} = ?$

iii) $V_1 = ?$, $V_2 = ?$, $V_3 = ?$

Connected in series

Solution:

For q_1:-

$$q_1 = C_1 V$$
$$= 10 \times 10^{-6} \times 250$$
$$q_1 = 2.5 \times 10^{-3} C$$
$$[q_2 = 2.5 mC]$$

For q_2:-

$$q_2 = C_2 V$$
$$= 50 \times 10^{-6} \times 250$$
$$q_2 = 12.5 \times 10^{-3} C$$
$$[q_2 = 12.5 mC]$$

For q_3:-

$$q_3 = C_3 V$$
$$= 25 \times 10^{-6} \times 250$$
$$q_3 = 6.25 \times 10^{-3} C$$
$$[q_3 = 6.25\, mC]$$

ii) $C_{Eq} = ?$ (Connected in parallel)

$$C_{Eq} = C_1 + C_2 + C_3$$
$$C_{Eq} = 10 + 50 + 25$$
$$[C_{Eq} = 85 \mu F]$$

$$Q = CV$$
$$V = \frac{Q}{C}$$

iii) In series

$$q = q_1 = q_2 = q_3$$

$$\therefore \frac{1}{C_{Eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

$$\frac{1}{C_{Eq}} = \frac{1}{10} + \frac{1}{50} + \frac{1}{25}$$
$$\frac{1}{C_{Eq}} = \frac{5+1+2}{50} = \frac{8}{50}$$

$$C_{Eq} = \frac{50}{8}$$

$[C_{Eq} = 6.25\,\mu F]$

$\therefore q = CV$

$q = 6.25 \times 10^{-6} \times 250$

$[q = 1.5625 \times 10^{-3} C = q_1 = q_2 = q_3]$

$\therefore V_1 = \frac{q}{C_1}$

$= \frac{1.5625 \times 10^{-3}}{10 \times 10^{-6}}$

$[V_1 = 156.25 V]$

$V_2 = \frac{q}{C_2}$

$= \frac{1.5625 \times 10^{-3}}{10 \times 10^{-6}}$

$[V_2 = 31.25 V]$

$V_3 = \frac{q}{C_3}$

$= \frac{1.5625 \times 10^{-3}}{25 \times 10^{-6}}$

$[V_3 = 62.5]$

Answer:

i) $q_1 = 2.5 mC$
$q_2 = 12.5 mC$
$q_3 = 6.25 mC$

ii) $C_{Eq} = 85 MF$

iii)
$V_1 = 156.25 V$
$V_2 = 31.25 V$
$V_3 = 62.5 V$

Question # 03:-

Data:

$V = 100 V$
$C_1 = 2 MF$
$C_2 = 3 MF$
$C_3 = 5 MF$
$q_1 = ?$, $q_2 = ?$, $q_3 = ?$
$V_1 = ?$, $V_2 = ?$, $V_3 = ?$

Solution:

C_2 & C_3 are connected in parallel, then their equivalent C' is given as:

$$C' = C_2 + C_3 = 3 + 5 = 8\,\mu F$$

Now C_1 & C' are connected in series:

$$\therefore C_{Eq} = \frac{C_1 \times C'}{C_1 + C'}$$

$$= \frac{2 \times 8}{2 + 8}$$

$$[C_{Eq} = 1.6\,\mu F]$$

$$\therefore q_{Eq} = C_{Eq} V$$

$$q_{Eq} = 1.6 \times 10^{-6} \times 100$$

$$[q_{Eq} = 1.6 \times 10^{-4}\,C]$$

Now C_1 & C' are in series. Both have same charge.

$\therefore q_1 = q' = q_{Eq} = 1.6 \times 10^{-4} C$

$$\boxed{q_1 = 1.6 \times 10^{-4} C}$$

$\therefore q' = Cq'$

$$V' = \frac{q'}{C'}$$

$$= \frac{1.6 \times 10^{-4}}{8}$$

$$\boxed{V' = 2 \times 10^{-5} V}$$